Study Protocol

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2. **List of abbreviations**

Abbreviation	Description
OMOP	Observation Medical Outcomes Partnership
OHDSI	Observation Health Data Sciences and Informatics
ICI	Immune Checkpoint Inhibitor
irAE	Immune-Related Adverse Events
RA	Rheumatoid Arthritis
SARD	Systemic Autoimmune Rheumatologic Disease
SjD	Sjogren's Disease
SLE	Systematic Lupus Erythematosus
SSc	Systemic Sclerosis

3. **Abstract**

Immune checkpoint inhibitors (ICIs) have transformed cancer therapy but are associated with immune-related adverse events (irAEs), including colitis, myocarditis, thyroiditis, and hepatitis. Emerging evidence suggests that patients with pre-existing systemic autoimmune rheumatologic diseases (SARDs) may be at increased risk for both autoimmune disease flares and irAEs when treated with ICIs. However, existing studies are limited by small sample sizes, retrospective single-center designs, and inconsistent phenotyping of both autoimmune disease and irAEs. Large-scale, standardized real-world evidence is needed to quantify irAE risk in this vulnerable population.

The objectives of this study are as follows:

1. To estimate the incidence of selected irAEs (colitis, myocarditis, thyroiditis, and hepatitis) among adult cancer patients with pre-existing or concurrent SARDs who newly initiate ICIs.
2. To estimate the incidence of the same irAEs among adult cancer patients without SARDs who newly initiate ICIs.
3. To assess whether the presence of a SARD is associated with a higher incidence of irAEs following ICI initiation.

4. **Amendments and Updates:** NA as of 12.18.2025

5. **Milestones:**
   - Finished protocol and updated the GitHub page

6. **Rationale and Background**

Immune checkpoint inhibitors (ICIs) have become key for treating multiple malignancies by enhancing immune responses. However, this immune activation can also lead to immune-related adverse events (irAEs), including colitis, myocarditis, thyroiditis, and hepatitis [1-4]. While irAEs are well described in the general oncology population, their incidence and risk factors remain incompletely understood and documented among patients with pre-existing systematic autoimmune rheumatologic diseases (SARDs).

Patients with SARDs are frequently excluded from clinical trials of ICIs due to concerns about autoimmune disease flares and heightened susceptibility to irAEs. As a result, the evidence base informing ICI safety in this population is limited to largely derived from small retrospective studies. Emerging research suggests that patients with preexisting autoimmune conditions may be at increased risk for both disease flares and new onset irAEs when treated with immune checkpoint inhibitors [5-10].

Existing research is further limited by inconsistent definitions of autoimmune disease and irAEs, limited sample sizes, and short follow-up. In particular, variation in phenotyping approaches has hindered comparability and generalizability. Therefore, OMOP enables standardized cohort definitions, reproducible outcome phenotyping, and consistent analytic approaches. By leveraging OMOP this study aims to generate robust estimates of irAE incidenc among ICI-treated patient with and without SARDs and to evaluate whether SARD status is associated with an increased risk of clinically important irAEs.

Understanding the incidence and relative risk of irAEs in patients with SARDs is critical for informing clinical decision-making, risk stratification, and patient counseling in oncology practice. This study seeks to fill a key evidence gap in providing scalable, generalizable real-world evidence on ICI safety in a historically underrepresented population.

7. **Study Objectives**
   a. Primary Hypotheses

      Among adult cancer patients newly initiating immune checkpoint inhibitors (ICIs), those with pre-existing or concurrent systemic autoimmune rheumatologic disease (SARD) experience a higher incidence of immune-related adverse events (irAES), specifically colitis, myocarditis, thyroiditis, and hepatitis, compared with patients without SARD.

   b. Secondary Hypotheses

      1. The incidence of irAEs (colitis, myocarditis, thyroiditis, and hepatitis) differs between ICI-treated patients with and without SARD.
      2. The magnitude of increase in irAE incidence varies by type of SARD.
      3. The association between SARD status and irAE incidence varies by cancer type and/or ICI class.

   c. Primary Objectives

      To compare the incidence of selected immune related adverse events (colitis, myocarditis, thyroiditis, and hepatitis) between adult cancer patients with pre-existing or concurrent system autoimmune rheumatologic disease and those without SARD who newly initiate immune checkpoint inhibitor therapy.

   d. Secondary Objectives

      1. To estimate the incidence of each individual irAE (colitis, myocarditis, thyroiditis, and hepatitis) among ICI-treated patients with SARD and without SARD.
      2. To assess whether irAE incidence differs by specific SARD subtype.
      3. To explore heterogeneity in irAE incidence across cancer types and immune checkpoint inhibitor classes.

8. **Research methods**

a. Study Design

This study will use a large-scale, observational, new-user cohort design implemented across multiple OHDSI network databases mapped to the OMOP Common Data Model. All analyses will be executed locally at each participating site using a common analytic package to ensure reproducibility and harmonization.

We will construct two parallel cohorts:

- Adults with cancer and preexisting or concurrent systemic autoimmune rheumatologic disease who newly initiate immune checkpoint inhibitor therapy.
- Adults with cancer who newly initiate immune checkpoint inhibitor therapy but do not have systemic autoimmune rheumatologic disease.

A new-user design will be applied so that the index date is the first recorded administration or prescription of an ICI agent.

b. Data Sources

The study will draw on multiple OHDSI network databases that include administrative claims or electronic health record data. All data are already mapped to the OMOP CDM, allowing standard cohort definitions, phenotypes, and analytical code to run consistently across sites.

c. Study population

Eligible participants are adults aged 18 years or older with a recorded diagnosis of cancer and evidence of ICI initiation. Systemic autoimmune rheumatologic diseases will be identified using validated concept sets, in combination with clinician specialty information and disease-modifying antirheumatic drug use in the year prior to ICI initiation. Patients must have at least one year of continuous observation prior to the index date to permit baseline covariate assessment.

Name	Concept Set Group	Atlas ID
Rheumatoid Arthritis (RA)	SARD (Condition)	80809
Lupus (SLE)	SARD (Condition)	257628
Systemic Sclerosis (Scleroderma, SSc)	SARD (Condition)	1134442
Sjogren's Disease (SjD)	SARD (Condition)	-
Dermatomyositis (DM)	SARD (Condition)	80182
Alveolar Sarcoma	Cancer (Condition)	4301648
Biliary	Cancer (Condition)	-
Cervical	Cancer (Condition)	4310575 (carcinoma of uterine cervix, invasive) 198984 (malignant tumor of cervix)
Classical Hodgkin's Lymphoma	Cancer (Condition)	42538850
Endometrial carcinoma	Cancer (Condition)	4110871
Endometrial dMMR	Cancer (Condition)	-
Esophageal	Cancer (Condition)	37204806 (undifferentiated)
Gastric	Cancer (Condition)	4155299 (carcinoma of stomach)
Hepatocellular	Cancer (Observation)	4101758
HNSCC	Cancer (Condition)	37396745 1076143
Malignant Melanoma	Cancer (Condition)	4162276
Merkel cell carcinoma	Cancer (Condition)	4100425
Mesothelioma (Sarcomatiod)	Cancer (Observation)	4161548
MSI-h colorectal cancer	Cancer (Condition)	42537577
MSI-h/MMR other	Cancer (Condition)	-
NSCLC, PD-L1 > 50%	Cancer (Condition)	4115276
NSCLC, PD-L1 : 1-50%	Cancer (Condition)	4115276
PMBCL	Cancer (Condition)	40481522
RCC	Cancer (Condition)	-
SCLC	Cancer (Condition)	4110591
Solid tumors, TMBh	Cancer (Clinical finding)	37163625
TNBC (triple negative breast cancer)	Cancer (Condition)	45768522
Urothelial (not cisplatinspecific)	Cancer (Condition)	37163178 (micropapillary urothelial carcinoma)
Urothelial (not cisplatin specific, PD-L1 > 10%)	Cancer (Condition)	37163178 (micropapillary urothelial carcinoma)

Urothelial cisplatin ineligible (50%)	Cancer (Condition)	37163178 (micropapillary urothelial carcinoma)
Pembrolizumab	ICI-PD1	45775965
Nivolumab	ICI-PD1	45892628
Cemiplimab	ICI-PD1	35200783
Dostarlimab	ICI-PD1	1536789
Retifanlimab	ICI-PD1	1302024
Toripalimab	ICI-PD1	747052
Atezolizumab	ICI-PDL1	42629079
Durvalumab	ICI-PDL1	1594034
Avelumab	ICI-PDL1	1593273
Ipilimumab	ICI-CTLA4	40238188
Tremelimumab	ICI-CTLA4	741851
Relatlimab (in combination with Nivolumab for Melanoma)	ICI-LAG3	1525060
Methotrexate	DMARD	1305058
Hydroxychloroquine	DMARD	1777087
Sulfasalazine	DMARD	964339
Leflunomide	DMARD	1101898
Azathioprine	DMARD	19014878
Mycophenolate	DMARD	19068900
Cyclophosphamide	DMARD	1310317
TNFi	DMARD	-
Abatacept	DMARD	1186987
Rituximab	DMARD	1314273
Tocilizumab	DMARD	40171288
Colitis	Outcome	4272488
Myocarditis	Outcome	314383 3656114 (due to autoimmune disease)
Thyroiditis (autoimmune)	Outcome	4281109
Hepatitis (autoimmune)	Outcome	200762

d. Exposures

The exposure of interest is systemic autoimmune rheumatologic disease status, defined as present or absent at the time of ICI initiation. The index date is the date of first ICI exposure. ICI categories will include PD-1 inhibitors, PD-L1 inhibitors, CTLA-4 inhibitors.

e. Outcomes

The primary outcomes are immune-related adverse events: colitis, myocarditis, thyroiditis, and hepatitis. Outcomes will be defined using validated or expert-derived phenotypes based on diagnosis codes, procedures, laboratory findings. Each IRAE will be evaluated separately.

f. <u>Covariates</u>

Baseline covariates assessed in the year prior to ICI initiation will include age, sex, cancer type, ICI class, comorbidity burden as measured by Charlson index, and prior immunosuppressive medication use. Covariates will support confounding control in multivariable models and assessment of cohort comparability.

9. Data Analysis Plan

a. <u>Calculation of time-at risk</u>

We will calculate separate time-at-risk for each IRAE outcome (i.e. colitis, myocarditis, thyroiditis and hepatitis). The first date of recorded ICI exposure will be used as the index date. Time at risk will begin from this date until the earliest occurrence of one of the following events:

- Occurrence of outcome of interest (specific IRAE)
- Discontinuation of ICI therapy: we define this as a gap of at least 60 days
- Death

b. <u>Model Specification</u>

Incidence rate for each IRAE will be calculated as the number of events divided by total person-time at risk, for each cohort. To compare the incidence of SARD and non-SARD cohort, we will use Cox proportional hazards models to estimate hazard ratio (HRs) and 95% confidence intervals. The primary exposure variable will be SARD status (binary: yes/no).

Models will be adjusted for specific baseline covariates, such as age, sex, cancer type, type of ICI (PD-1, PD-L1, CTL-2 or combination therapy), baseline immunosuppressive medication and Charlson comorbidity index.

c. Pooling effect estimates across database

Each database in our network study will conduct the analysis with our code. We will pool the hazard ratios together utilizing a random-effects meta-analysis to account for between-database heterogeneity. Statistical heterogeneity will be assessed using Cochran's Q test. We will report all individual hazard ratios in the supplementary information of our publications.

d. Analysis to perform

1) Descriptive statistics analysis: baseline characteristics of the SARD and non-SARD cohorts.
2) Incidence analysis: incidence rates of each IRAE in SARD and non-SARD cohorts
3) Comparative analysis: Hazard rates for each IRAE incidence between cohorts. This will be outcome specific.
4) Sensitivity analysis: restricting SARD definitions to validated phenotypes only and varying the treatment discontinuation gap.

e. Output

We will generate all outputs using OHDSI-compliant analytical tools to ensure reproducibility. Outputs will include:
- Cohort flow diagram
- Tables of baseline characteristics by cohort and database
- Incidence rates for each IRAE
- Data-base specific and pooled hazard ratios

f. Evidence Evaluation

We will interpret the results within the context of known limitations of observational data, including potential cofounding, outcome misclassification, and incomplete capture of laboratory-based phenotypes.

10. **Strengths and Limitations of the Research Methods**

This large scale, multi-database cohort study that utilizes OHDSI's federated framework, enabling consistent execution of standardized analyses across different data sources. The inclusion of multiple databases will increase our statistical power, generalizability, and strength of our findings. However, residual confounding effects do remain a limitation. Certain clinically relevant covariates, such as disease severity and detailed treatment intent, may be incompletely captured or unavailable within OMOP.

11. Protection of Human Subjects

All data is de-identified prior to researcher access. No direct contact with human subjects will occur, and no intervention will be performed as part of this study. We will be utilizing data that has already been collected by the clinicians. Each participating data partner will need to obtain regulatory approval through an IRB form as required by their institution. No individual patient information will be published with this study, only aggregated from the various participating data partners.

12. Management and Reporting of Adverse Events and Adverse Reactions

As our data will be anonymized, we will not be able to report adverse reactions and events back to patients and institutions.

13. Plans for Disseminating and Communicating Study Results

Results will be shared with academic presentations and publications.

14. References

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